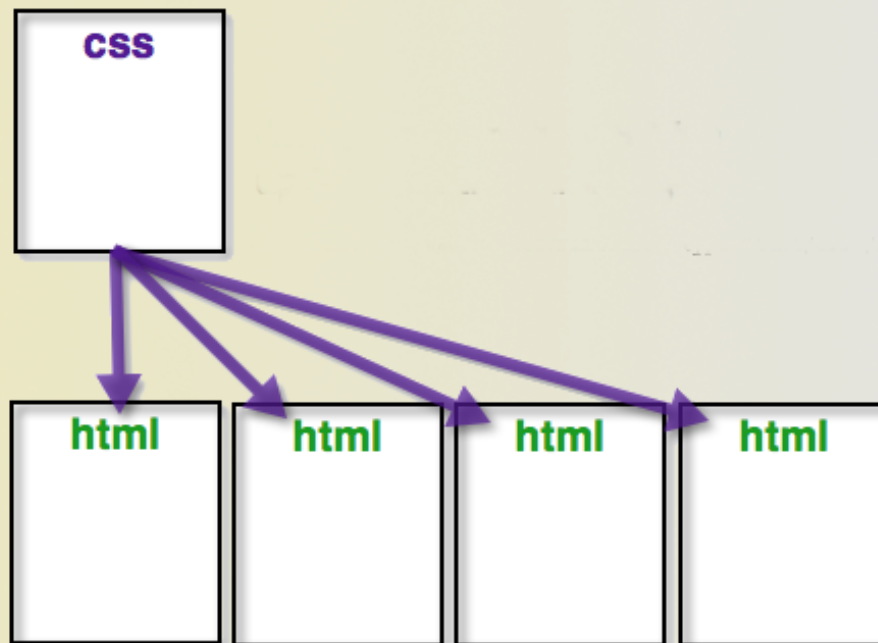


3 Ways to Use CSS

External, Embedded, In-line



The Three Ways

External

External style sheets hold CSS used by all the pages on a web site.

The CSS in an external style sheet are common to every page. They are general.

Internal or Embedded

Embedded CSS is added to a web page in the <head> element of the page using the <style> element.

Embedded CSS is specific to that page only. It is used to style the HTML that page only.

In-line

In-line CSS works on the element level. This CSS only affects the specific tag. It is very specific.

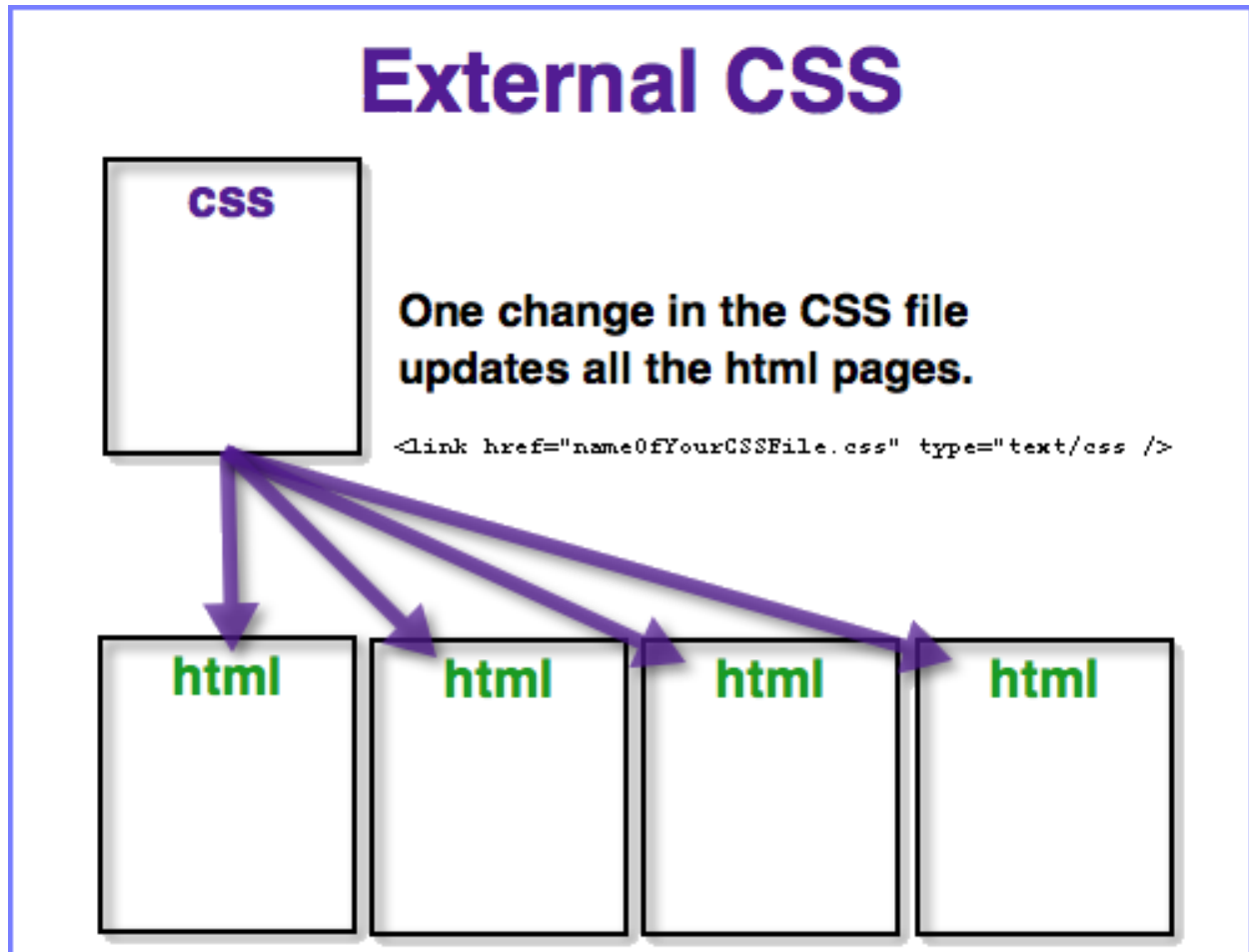
Lab Exercise:

1. Set up a project folder named: externalCSS
2. Inside this folder set up three HTML pages. Name them: index.html, contact.html, product.html
 - Include a <title>, <h1>, and a few <p> of text on each page.
 - Make the text different different on each page, especially the <title> and <h1>
3. Also inside this folder create an external CSS file
 - Write a single selector for <h1>
 - background: gray;
 - border: 1px solid blue;
 - color: blue;
 - margin: 15px;
 - padding: 10px;

- Make certain there is no HTML code on the page
 - Comments should use `/* */`
 - No `<` on the page! Use CTRL f to double check for any.

You will be using these four files to explore the three ways to use CSS.

External



External CSS selectors are stored in a separate, stand-alone CSS file. Each HTML file on the site connects to the external CSS file using the `<link>` element:

```
<link href="nameOfYourCSS.css" rel="stylesheet" type="text/css" />
```

This `<link>` element must be inside of the `<head>` element.

In essence what the browser does, as it loads the page, is insert all the code from the external file into the temporary memory for that particular page. As the browser parses through the HTML code it uses this temporary memory to

determine how to style each element.

Advantages of external CSS

- Change CSS selector in one file and the results cascade to all pages using that external file.
- A great place to keep CSS selectors that are common to the all HTML pages.
- Gives the site a consistent look and feel because every page is being styled the same.

Disadvantages of external CSS

- Adds complexity. Not useful for single-page web sites
- Often a server will not allow external files. Many learning platforms such as D2L and Blackboard are like this.
- No customization. All pages are styled the same.

Things to Check

- Each CSS selector must have a set of curly braces { }. The rules for that selector will go inside the curly braces.

For example, here is an h1 selector with five rules:

```
h1 {
background:      pink;
border:         1px solid #000080; /* dark blue */
color:          #000080;
margin:         15px;
padding:        10px;
}
```

- Make certain there is no HTML code inside the CSS file. Use CTRL f to "find" any HTML by looking for: <

Notice that the selector for <h1> in CSS looks like this: h1
- Make sure the external CSS file is in the same folder as the HTML files.
- Make sure you spell the CSS filename correctly in the <link /> element. Developers often name the CSS file the same as the website. You can

copy and paste the name so you are absolutely certain that it matches.

Lab Exercise

1. Using the `<link />` element connect each of the web pages you wrote with the external CSS.
2. Be smart! Get the external linked with one page, double check it, and then copy and paste the `<link />` into the `<head>` section of the other two pages.
3. View each page and make certain the `<h1>` is being styled as expected.

Internal/Embedded

Internal/Embedded CSS
Supports the html on a single page.

```
<head>
<style type="text/css">
h1 {
padding: 10px;
margin: 15px;
}
</style>
</head>
<body>
<h1>My Heading</h1>

</body>
</html>
```

**This CSS styles
this HTML code.**



Internal or embedded CSS selectors are added to an individual page using the `<style>` element inside the `<head>` section of the page.

As the browser loads the page it takes the information in the `<head>` element and stores it in temporary memory. As the browser reads in the remainder of the page it uses this temporary memory to determine how to style each element.

What if the same selector is in the external style sheet? Who wins?

Internal/embedded CSS on a page is more specific. It has precedence or

overrides the CSS code from an external style sheet.

General ---> Specific

External CSS --> Internal/Embedded CSS

Advantages of embedded CSS

- CSS and HTML are all on the same page.
- You have more specific control over the styling of the page.
- A great way to test CSS code before moving it out to an external file where it will affect the entire site.
- The CSS is part of the HTML file. Can be used on web sites and email servers that don't understand external files.

Disadvantages of embedded CSS

- If the CSS is the same for several pages you end up with lots of redundant code.
- To make changes across several pages you have to open up all the files (and then FTP or copy them up to the web server.)

Things to Check

- Make certain the CSS is inside of the `<style type="text/css">` element inside the `<head>`.
- Check spelling and punctuation:
 - Every selector must have curly braces { } around the list of declarations (the property, styles, and comments)
 - UPPER/lower case matter.
 - Properties end with colon ":" and styles end with semi-colons ";"

Lab Exercise

1. Write a new selector for the `<h1>` element on the index.html page using this code:

```
h1 {  
background:  green;  
color:      white;  
font-family: helvetica, arial, sans-serif;  
font-size:  2em;  
}
```

2. View all the pages.

- Did the h1 on the index.html change? (It should have.)
- Did the border change? The margins? The padding? Where did index.html get these CSS styles from?
- Did the font-family or font-size change on any of the other pages?
- Which is more general? The external or the internal/embedded? Which is boss?

In-line

Inline CSS

Affects only the current element.

```

```

These properties only affect this element.

Multiple changes have to be made everywhere an inline style is used.

Inline CSS is added to individual HTML elements using the `style=" "` attribute.

For example, to style a specific `<h1>` attribute you could use this code:

```
<h1 style="color:purple; border:5px purple solid; font-  
family:'Times Roman', Courier, serif; background:#CECECE;">  
This is the Heading Text  
</h1>
```

If a style requires quotes then single quotes should be used so there isn't a conflict with the double quotes used for the `style=" "` attribute itself. An example of this would be using a font name that has two words such as 'Times Roman'.

What if the same selector is in the external style sheet or the embedded styles at the top of the page? Who wins?

Inline styles are always the most specific. They override both the internal/

embedded CSS as well as the generalized CSS in an external file..

General ---> Specific

External CSS --> Internal/Embedded CSS --> Inline CSS

Advantages of inline CSS

- Great choice when you need to style one object on the page a certain way, once.
- Complete control of individual HTML elements on a page.
- The CSS is part of the HTML file. Can be used on web sites and email servers that don't understand external files.

Disadvantages of external CSS

- High maintenance. If you have a lot of inline styles you will need to track each of them down to make changes across the site.
- Potential for lots of duplication. If the same CSS is used for more than one HTML element it should either be embedded on the page or put in an external CSS file.
- Often difficult to read because everything is mashed together with no spaces.

Things to Check

- All the inline CSS must be inside the double quotes of the style=" " attribute
- The CSS code itself should not use any double quotes (they conflict with the attribute quotes.)
- Make sure the punctuation is correct. Each property must end with a colon ":" and each style must end with a semi-colon ";".

Lab Exercise

1. On the contact.html page use inline CSS to style the text of the <h1>

element.

```
<h1 style="color:white; background: green;  
border:blue;padding:5px 20px 5px 30px;">  
This is the Heading Text  
</h1>
```

2. Add another `<h1>` on the `contact.html` page (without any inline styling).

3. View all the pages.

- Did the `h1` on the `index.html` change? How about on `product.html`? Did it change on `contact.html`?
- Did the border change? The margins? The padding?
- Did the inline style change any of the other `<h1>` elements? How about the second one you added to the `contact.html` page?
- Which is the most general? The external CSS, internal/embedded, or inline?
- Which is most specific? The external CSS, the internal/embedded, or the inline?

Who rules? (precedence)

Always think general to specific.

General - Add to the external CSS file.

When creating a web site put the majority of your CSS selectors in an external style sheet and link to that files with all of your pages.

More Specific: Use embedded CSS

Then, as you work with individual pages that have unique styling, add those styles in the <head> section of that particular page.

Or, if you are in a situation where an external style sheet is not allowed, then embed the CSS styling.

Very Specific: Use inline CSS

On the rare occasion that you need to style a very specific element on a page use inline styling. This method should be used very infrequently because the maintenance is very high having inline code everywhere.

Summary

Three ways to implement CSS

Here are three methods. What are the advantages and disadvantages of each?

1. External - Implementing CSS across multiple pages.
 - What is the HTML code that is used to link the external CSS file with a web page?
2. Internal/Embedded - Using CSS to style elements on a single page
3. In-line - Implementing CSS for a single element on a page.

Which CSS style has precedence?

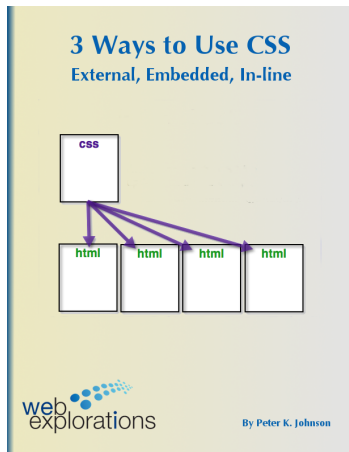
Which method should you use in which situation?

Credits

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